Safety Instructions



This section is intended to help you operate your Uniloy machine safely and productively.

All reasonable precautions were taken to assure the safety of personnel using your machine. Personnel must practice safe work procedures at all times on and around the machines. Safety and Instruction signs are placed to alert personnel to hazards and precautions to use.

Your machine has a system of safety devices which help assure safe machine operation.

⚠WARNING

All machines must be operated with the prescribed safety devices (guards, interlocks, signs) properly positioned and in good working order, as originally designed and installed.

The absence of these safety devices from a machine that appears in any of our manuals, training programs, or sales media occurs for display purposes only.

⚠WARNING

You must read and understand these precautions completely before operating, setting up, running, or performing maintenance on the machine. Failure to follow this instruction may result in serious personal injury and/or machine damage.

These safety precautions are to be used as a guide to supplement the following:

- All other information pertaining to the machine.
- Local, plant and shop safety rules and codes.
- Federal and state safety laws and regulations.

Note: *Emphasis is placed on the latest edition of the Occupational Safety and Health Standards, which is available from the Department of Labor, Washington, D.C. These standards (found in Part 1910, Title 29 of the Code of Federal Regulations) contain the current, general industry occupational safety and health regulations set forth by federal legislation. Also applicable are standards available from the American National Standards Institute. For example, A.N.S.I. B 151.1 is the Safety Standard for horizontal injection molding machines.*

⚠WARNING

Addition to this machine of ancillary equipment such as robots, part removers, and mold loaders can create hazards producing serious injury or death. Proper safeguarding systems shall be provided to prevent access to these hazards. Follow the manufacturer's instructions for proper installation, operation, and maintenance. Failure to follow these instructions may result in serious personal injury and/or machine damage.

⚠WARNING

Request from your resin supplier a Material Safety Data Sheet (MSDS) to determine which, if any, health exposure hazards are present during processing. Failure to follow these instructions may result in serious personal injury and/or machine damage.

Personal Safety Attitudes and Actions

All reasonable precautions must be taken to assure the safety of personnel using the machine. Only qualified, trained personnel may operate the machine.

In order to avoid accidents, you must place a high priority on safety, and your thoughts and actions must reflect that priority. Know and respect your machinery. Read and practice the prescribed safety procedures. Make sure that everyone who works for you, with you, or near you, fully understands and follows these safety precautions and procedures when operating or servicing the machine.

⚠WARNING

Read the Machine Installation section of this manual, and resolve any questions you may have, before attempting to move or install the machine. Obey all safety precautions. Failure to follow this instruction may result in serious personal injury and/or machine damage.

Always be mentally alert on the job, and stay in good physical condition. A person who is not alert runs a greater risk of getting injured.

Never operate or service this machine if affected by alcohol, drugs, or other substances or conditions which decrease alertness or judgment.

Avoid sudden movements, loud noises, etc., which can distract others who are working near you. Be alert to conditions in your working area and throughout the plant. Observe 'No Smoking', 'High Voltage', and other warning signs.

⚠WARNING

Do not climb on the machine. Climbing on the machine could cause a machine malfunction, and may result in injury if you slip or are bumped by the machine. Also, do not stand on any part of the machine and reach into the mold area while the machine is operating. Failure to follow this instruction may result in serious personal injury and/or machine damage.

Every tool should be used only for its intended purpose. Check tools and equipment frequently. Never use worn tools, frayed extension cords, etc. Make sure that all electrical equipment is properly grounded. Short cuts and safety are not compatible.

During machine operation, be attentive to the molding process. Excessive vibration, unusual sounds, etc., can indicate problems that require your immediate attention.

Place oily rags in a covered metal container, and throw all trash and scrap in the proper waste containers. Nails, broken glass and chips are particularly dangerous. Always use a broom and pan to pick up debris. Never pick up debris with your bare hands, or serious cuts may result. Remove nails from lumber or containers that may be handled by you or others.

⚠WARNING

The proper clothing for the job must be worn at all times. Always wear approved eye or face protection (glasses, shields, etc.), and keep them clean. Wearing safety toe shoes with skid-proof soles can help to keep you from slipping and falling. Wear a safety hat. Keep your protective equipment in good condition and be sure it meets or exceeds any required or recommended standards. Failure to follow these instructions may result in serious personal injury.

⚠WARNING

Accidents can occur due to clothing or other articles becoming entangled in moving machine components. Neckties, scarves, loose hanging clothing, and jewelry such as watches, rings, or necklaces must not be worn around moving machinery. Wear a short sleeve shirt, or roll your sleeves up past your elbow, and keep your shirt tucked in. If you have long hair, restrain it with a cap or a net. Wear gloves only when essential. Never wear gloves when they can become entangled in the machine. Failure to follow these instructions may result in serious personal injury.

When improperly used, compressed air can be dangerous and can result in serious injury. Never use compressed air for cleaning the machine, work area, workbench, tools, workpieces, measuring instruments, or clothes. When cleaning, use some form of vacuum device instead of compressed air. Make sure that air blasts from air-operated tools do not carry chips or dirt toward you or others. Never blow air toward other people or toward the machine, since the air nozzle itself may contain small chips.

Keep your machine in good operating condition at all times. It should be inspected regularly for parts that show wear or abuse. Worn parts should be replaced with authorized replacement parts only. Report any unusual sounds, smoke, heat, damaged components, etc. to your supervisor.

Work Area Safety

Make sure your work area is free of hazardous obstructions, and be aware of protruding machine members while working around the machine. Keep materials and equipment out of the aisles. Equipment should be returned to its proper storage place after use. Keep your workbench neat and clean. Keep tools and workpieces in order.

⚠WARNING

Do not leave foreign articles laying on the machine. They could cause injury by distracting or hitting someone. They could also cause the machine to malfunction by shorting an electrical circuit, or by jamming the machine. Failure to follow this instruction may result in serious personal injury and/or machine damage.

If work platforms are used around the machine, make sure they are sturdy and they include anti-slip surfaces. Report worn or broken flooring, stair treads, or handrails, or any other unsafe conditions, to your supervisor. When climbing, make sure ladders and scaffolds are well built and in good condition. Never climb on stock, machines, or makeshift ‘ladders’ built on skids, tote pans, boxes, etc.

Have proper lighting around the machine, electrical equipment and cabinets. A poorly lit work area can lead to eye strain or an accident. Light should not be directed into your eyes.

Keep working space around electrical equipment clear of obstructions, and arranged in such a way to give authorized personnel convenient access to all areas requiring attention.

Avoid grinding operations near the machine. Abrasive dust causes undue wear, inaccuracies, and possible failure of affected parts.

⚠WARNING

Always keep your working area clean. Dirty work areas with such hazards as oil or water on the floor may cause someone to fall to the floor, into the machine, or onto another object. If spillage or leakage of any material occurs, clean up the material immediately. Failure to follow this instruction may result in serious personal injury.

Installation and Relocation Safety

Before lifting the machine, consult this manual or Uniloy for proper methods and procedures. Drain all oil from the reservoir(s) before lifting the machine.

An electrician must read and understand the electrical schematics prior to connecting the machine to the power source. After connecting the machine, test all aspects of the electrical system for proper functioning. Always make sure the machine is grounded properly. Place all selector switches in their 'Off' or neutral (disengaged) position. The doors of the main electrical cabinet must be closed and the main disconnect switch must be in the 'Off' position after the power source connection is complete.

Always lock the main disconnect device in the 'Off' position if the machine is left unattended, unless the machine is in an automatic production cycle.

When the machine is installed, be sure that the motor(s) rotate in the proper indicated direction.

Setup and Operation Safety

Familiarize yourself with the machine before attempting any setup, operation, or maintenance work. Respect the machine's power and capability. Know where all stop buttons are located in case of an emergency, and be familiar with all warning lights or alarm buzzers that warn of a malfunction. Know the controls and all external moving parts of the machine. Read and understand all sections of this manual, and operate the machine only as set forth in the manual.

⚠WARNING

When the machine first arrives at your plant, or when it is relocated, certain safety precautions must be observed. Do not attempt to lift, move, or install the machine before consulting the Machine Installation section of this manual for proper methods and procedures. Be sure your lifting equipment is rated for safe lifting well above the machine weight. Resolve any questions about lifting, moving, or installing the machine before beginning any of these procedures. Failure to follow these instructions may result in serious personal injury and/or machine damage.

Only qualified personnel, instructed in safety and all machine functions, should be entrusted to operate this machine. Never change the machine settings, or attempt to repair the machine, unless you are trained and authorized to do so.

⚠WARNING

Do not attempt to perform any setup, operation, or maintenance work on the machine until you read and understand all sections of this manual. Failure to follow this instruction may result in serious personal injury and/or machine damage.

All safety guards, shields, barriers, covers, and protective devices must be connected or in place before operating the machine. All safety features, disengagements, and interlocks must be in place and functioning correctly prior to operation of this equipment. Never bypass or wire around any safety device.

When the process requires plasticizing at elevated barrel temperatures, begin operation at lower temperatures and gradually increase the barrel heat settings while the material is flowing.

⚠WARNING

When adding raw material to the machine, do so in a safe and stable manner. Do not climb or stand on the machine. Failure to follow this instruction may result in serious personal injury and/or machine damage.

⚠WARNING

Do not reach into the clamp area until all machine motion has stopped. Under no circumstances should a person climb the structure of the machine to reach into the clamp area from the top. Also, never reach in under a gate or a guard. Never reach over, under, or around any machine gates or guards while the machine is operating. Failure to follow these instructions may result in serious personal injury.

⚠WARNING

Shut off power to the machine if you leave the operating area. Never leave a machine running unattended. Always turn off the main disconnect switch before attempting to clean the machine at the end of the working day, or if guards or covers are removed that would expose hazardous areas. Failure to follow these instructions may result in serious personal injury and/or machine damage.

⚠CAUTION

Do not operate the machine if the oil temperature rises above 130°F (54°C). Stop the machine and investigate the cause. A clogged heat exchanger or coolant line, or a pump overload, may be at fault. Failure to follow these instructions may result in machine damage.

⚠WARNING

Attend to all oil leaks immediately. Leaks in the high pressure oil lines expand rapidly. Leaks in the low pressure lines can allow air to enter the system. Failure to follow this instruction may result in serious personal injury and/or machine damage.

⚠WARNING

Operating and maintenance personnel must carefully read, understand, and fully comply with, all machine mounted warning and instruction plates. Do not remove, paint over, alter, or deface these plates. Replace all plates which become unreadable. Replacement plates are available in any language upon request from Uniloy, Inc. Failure to follow these instructions may result in serious personal injury and/or machine damage.

Keep all electrical equipment in safe and proper working condition. Defective equipment should either be put in good order, or be permanently disconnected. Thoroughly inspect electrical equipment before using it to make sure it is in good operating condition.

⚠WARNING

If one or more guards must be removed or altered to facilitate tooling installation, or for any other reason, the user must provide additional guarding of sufficient size to prevent access to moving machine members when the machine is operated. This warning applies to any guard alteration or use; for example, whenever hazards are created in the use of dies or when die sets extend beyond the operating envelope of the machine. Failure to follow these instructions may result in serious personal injury and/or machine damage.

⚠WARNING

If your machine has hydraulic mold clamps, do not open the mold until all of the hydraulic mold clamps are fully locked and are so indicated. Do not remove safety catch plates from the mold clamp bases. Failure to follow this instruction may result in serious personal injury and/or machine damage.

Maintenance Safety

Before Starting

⚠WARNING

Do not attempt to perform maintenance on this machine until you read, know and practice all the safety instructions in this manual. Read and comply with all machine mounted warning plates. Know and practice safety rules set forth by your company. Failure to follow these instructions may result in serious personal injury and/or machine damage.

Never attempt to make any repairs for which you are not properly trained. Only qualified service and maintenance personnel should perform maintenance and repair work on this machine. Always consult the manual before attempting any service or repair work, and if you have any questions, contact a Uniloy office for further information. Use only Uniloy replacement parts - others may impair the general safety and operation of the machine. Warning or danger signs should be placed conspicuously around the machine before any maintenance or service work is begun.

When Working on Electrical Equipment

⚠WARNING

Before working on any electrical circuits, lock the machine's main disconnect switch in the 'Off' position. Do not work on the machine with electrical power on. Any maintenance work that must be performed with electrical power on should only be performed by a Uniloy field service representative. The employer, and subsequent transferees, must determine that any other person performing work with electrical power on is trained and technically qualified to do so. Failure to follow these instructions completely may result in serious personal injury and/or machine damage.

Before removing or opening any electrical enclosure, cover, plate, or door, make sure the main disconnect switch is turned off. If any tool is required to remove a guard, cover, bracket, or any basic part of this machine or control, lock the main disconnect switch in the 'Off' position. Also, post a sign at the disconnect switch indicating that maintenance is being performed.

Even with all electrical power disconnected from the machine, there is high voltage stored in some components of the control (such as capacitors). Also, some residual voltage is present for a time. Only qualified electricians (electrical service technicians who have been trained and designated to work on the control circuitry) should be allowed to work on the control. Before servicing the control, all stored voltage and all residual voltage must be drained or discharged from the affected components and circuits.

Arc flash can cause electrocution and burn hazards, so proper personal protective equipment (PPE) should be used. For more detailed information, at a minimum you should refer to NFPA 70, NFPA 70E and NFPA 79.

When removing electrical equipment, place numbered or labeled tags on any wires which are not marked. If wiring is replaced, be sure it is of the same type, length and size. Also, be sure that it has the same load carrying capacity. Close, and securely fasten, all guards, shields, covers, plates and doors before power is reconnected to the machine.

Working space around electrical equipment must be clear of obstructions. Provide adequate illumination to allow for proper operation and maintenance.

⚠WARNING

National Fire Protection Association standards require that signals entering or leaving this unit from or to any external source over 30 volts, and not disconnected when the main disconnect switch is turned off, be identified by using conductors with yellow insulation (or yellow sleeving or marking on such wires if multi-conductor cables or other special wiring methods are used). It is the responsibility of the owner/user to also disconnect these circuits identified by conductors with yellow insulation or yellow markings before performing maintenance. Failure to follow this instruction may result in serious personal injury and/or machine damage.

The nearest Uniloy regional office must be notified before any changes are made to the control software. Software changes must be made by qualified personnel, using the supplied or approved methods and equipment. Immediately after the control is re-programmed, the following conditions must be checked.

a. Any change to the programming must include all previous safety related devices, and any new safety devices or methods of safe machine operation which may be necessary.

b. All safety-related devices must be inspected and tested to make sure they are working properly.

c. Safety guards, doors, shields, limit switches, etc., must be interlocked so the machine will not operate if they are opened, actuated, tripped, removed, or damaged.

It is mandatory that all wiring diagrams, programming sheets, programming tapes, and other documentation be updated when changes are made to the control program, or to any electrical or electronic circuit hardware. All personnel who come in contact with the machine must be immediately alerted to any and all changes that are made.

It is the responsibility of the owner/user to add any additional protective components that may be needed whenever changes or variations are made to any of the machine components or operational characteristics.

Regularly inspect all machine safety devices to make sure they are functioning properly. Replace all non-functioning components immediately, before operating the machine. All safety-related components such as guards, shields, barriers, covers and doors must be in place before and during operation of this machine.

⚠WARNING

Before performing any work on the machine, and before removing or opening any electrical or mechanical guard, cover, plate, or door, be sure the main disconnect switch is locked in the 'Off' position. Also, post a sign at the disconnect switch to indicate that maintenance is being performed. Whenever work is to be performed in any area away from the disconnect switch, and the disconnect is not locked, tag it and all start buttons with a 'Warning' or 'Do Not Start' tag. Adequate precautions, such as locks on circuit breakers, warning notices, or other equally effective means, should be taken to prevent electrical equipment from being energized when work is being performed. Failure to follow these instructions may result in serious personal injury.

Pilot and Relief Valves

⚠WARNING

All pilot and most relief valve pressure settings are established at the factory. Changing these settings from the recommended ones could result in damage to the machine and voiding of the warranty. If it is necessary to adjust any pressure setting, consult the machine prints or contact the Uniloy field service department. Failure to follow this instruction may result in machine damage.

Working on Components That Can Fall

⚠WARNING

When working on any component, always be sure it is safely positioned - for example, on a sturdy workbench. Never work on a component while it is hanging from a crane or any other lifting mechanism. Failure to follow these instructions may result in serious personal injury and/or machine damage.

Forcing and Hammering Parts

Never hammer or force any part of the machine to make it work. If it is necessary to strike the work-piece or fixture for adjustment, use a soft hammer.

Before Reconnecting Power

⚠WARNING

Orange identifying markings on covers, guards, etc. indicate that they are to be remounted immediately after adjustments have been made to the components under them. Make sure all guards, covers, plates and doors are closed and securely fastened before power is reconnected. Failure to follow this instruction may result in serious personal injury and/or machine damage.

Ventilation Safety

Fumes, dust and gases may be produced in a plastics processing environment. Adequate ventilation must be supplied. It is recommended that the construction, installation, inspection and maintenance of exhaust systems conform to the requirements of A.N.S.I. Z9.2, 'Design and Operation of Local Exhaust System', and also to any recommendations from the material supplier.

Materials Safety

Various materials may be used with this machine. Before using, mixing, or diluting materials in the machine, contact the manufacturer or authorized supplier of the material to determine whether or not the material is suitable for the intended application, and request a Material Safety Data Sheet (MSDS) from the material manufacturer.

Lifting and Carrying

Contact supervision if you have any questions or are not sure about the proper procedures for lifting and carrying. Improper lifting or carrying can result in serious injury - especially back injury. The following suggestions may help you to avoid injury:

- Before lifting or carrying an object, determine the weight and size by referring to such things as tags, shipping data, labels, marked information, or manuals. Inspect the load for nails or other protrusions that might cause injury. Decide whether it can be moved by one person, or several, or whether a device such as a crane or a fork lift is required. Get any necessary help before proceeding.
- If in doubt as to the size or type of lifting equipment, method, and procedures for lifting, contact Uniloy before proceeding to lift the machine or its components.
- Before inserting an eyebolt, be certain that both the eyebolt and the hole have the same size and type threads. Do not attempt to use a metric thread bolt in an English thread hole or vice versa. To attain safe working loads, at least 90% of the threaded portion of a standard forged eyebolt must be engaged.

Use power hoists, or other mechanical lifting and carrying devices, for heavy objects that exceed weights set forth by safety standards, or your own physical limitations. Objects that are bulky or hard to handle should also be lifted or carried with the aid of mechanical devices. This can help you avoid injury. Use hook-up methods recommended by your safety department, and know the signals for safely directing a crane operator.

Lifting and Carrying By Hand

Make sure your footing is secure. Crouch as close to the object as practical. Get a good grip on the object. Lift slowly by straightening your legs. (Keep your back relatively straight. Your arm and leg muscles, not your back, should do the work.) Keep the load close to your body as you come up.

Lift the load to the carrying position. If it is necessary to change your direction when in the upright position, turn your body with changes of foot positions. Be careful not to twist your body.

If you set the load on a bench or table, place it on the edge to make the table take part of the weight. Then push the load forward using the arms, or if necessary, part of the body.

If you must lift the object higher than your waist, first lift the load waist high and rest it on a support (as described in the preceding paragraph). Then change your grip and bend your knees again to give added leg muscle power for the final lift.

When carrying an object, always have a clear vision over the load. Do not try to change the load's position, or adjust it, while you are in motion. If the load interferes with normal walking, get help.

When putting the load down to the floor surface from a waist high carrying position, bend the knees, and, with a straight back and the load close to the body, lower the object with the arm and leg muscles.

Lifting and Carrying By Mechanical Device

Use power hoists, or other mechanical lifting and carrying devices, for heavy objects that exceed weights set forth by safety standards, or your own physical limitations. Objects that are bulky or hard to handle should also be lifted or carried with the aid of mechanical devices, rather than by hand. This can help you avoid injury. Use hook-up methods recommended by your safety department, and know the signals for safely directing a crane operator.

⚠WARNING

Regularly inspect slings, chains, hoists and other lifting devices. For frequency of inspection, refer to the Williams-Steiger Occupational Safety and Health Act, 29 CFR 1910.184, and any instructions applicable to the equipment. Any unsafe equipment should be properly repaired or immediately discarded. Failure to follow these instructions may result in serious personal injury and/or machine damage.

⚠WARNING

Cranes, hoists, slings, eyebolts, and other lifting equipment have safety rated capacities which should never be exceeded. Before using any lifting equipment, make sure the equipment is adequate for the load and application. Refer to standards and instructions applicable to any lifting equipment you use. (For example ASME B18.15, published by the American Society of Mechanical Engineers, United Engineering Center, New York, contains information concerning safe lifting loads for different size eyebolts, for various angles of lift, and for varying applications.) Failure to follow this instruction may result in serious personal injury and/or machine damage.

⚠WARNING

Never place any part of your body under a suspended load or move a suspended load over any part of another person's body. Be certain that you have a safe spot for depositing the load before lifting it. Failure to follow these instructions may result in serious personal injury and/or machine damage.

There are many other safety suggestions that could be made. If there is a question regarding safety, ask a trained person in your company, or call the Uniloy representative in your area.

Color Coding System

When color coding is used to indicate hazardous parts of machines, colors shall conform to A.N.S.I. Z535.1. The parts of that standard which apply to your machine are included below. Since your safety, as well as the maximum productivity of the machine, is of the utmost concern, everyone who comes in contact with the machine should read and understand the color coding explanation that follows.

Safety Red shall be the color for the identification of Danger and Stop. The following are some common examples of applications where the color Safety Red may be used:

- The background color of the signal word panel for Danger safety signs, labels and tags.
- Emergency stop bars on machines.
- Stop buttons or electrical switches used for emergency stopping of machinery.

Safety Orange shall be used for the identification of an intermediate level of hazard (i.e. Warning). If a color is to be used to identify the hazardous parts of machines, Safety Orange shall be used. The following are some common examples of applications where the color Safety Orange may be used:

- The background color of the signal word panel for Warning safety signs, labels and tags.
- Marking hazardous parts of machines which may cut, crush, or otherwise injure.

Safety Yellow shall be the color for the identification of Caution. The following are some common examples of applications where the color Safety Yellow may be used:

- The background color of the signal word panel for Caution safety signs, labels and tags.
- Marking physical hazards which might result in: striking against, stumbling, falling, tripping, or being caught in-between.

When surfaces of the machine have been color coded for safety reasons, these colors should not be changed.

Lockout / Tagout Procedure

Effective January 8, 1990, O.S.H.A. has designated the need for a ‘positive, lockable’ means to remove all energy sources from equipment prior to service or maintenance. The electrical power to your machine can be locked out at the main disconnect switch, which is normally located on the electrical cabinet. When this is done, residual energy remains for some time in the capacitors associated with the electrical system. This residual energy is automatically depleted by features built into the machine. After locking out the main disconnect switch, wait at least 60 seconds before beginning any maintenance procedures. This allows the residual energy to diminish. (If a machine-mounted plate indicates that you should wait longer that this, wait the recommended period of time before beginning any maintenance work.)

Whenever you need to perform maintenance on the machine, or whenever you need to shut it down for any other reason, a lockout procedure must be followed. Your employer is required by O.S.H.A. to develop a written lockout/tagout procedure for this machine. The following items should be considered in developing this procedure.

a. Notify everyone who normally operates, sets up, or performs maintenance on the machine that it will be shut down.

b. Turn off the electric motor.

c. Turn off the main electrical disconnect switch.

d. Lock the main disconnect switch in the 'Off' position, and place a tag on the switch to indicate that work is being performed on the machine.

e. If there is any auxiliary equipment associated with the machine, make sure the main electrical disconnect switch is also turned off for each piece of auxiliary equipment. Then lock each disconnect switch in the 'Off' position, and tag each switch to indicate that work is being performed on the machine.

f. Lock the air and water supply valves to make sure no air or water can be supplied to the machine.

g. Verify that no sources of residual energy (capacitors, suspended machine components, etc.) are present on the machine or any piece of auxiliary equipment. If any such energy sources are located, make sure they are neutralized. If necessary, manually discharge hydraulic pressure and capacitor voltage from charged components. Also, block all suspended or spring loaded machine parts to prevent movement.

h. Verify that electrical power has been disconnected from the molding machine, and from any auxiliary equipment, by trying to energize the machine and any auxiliaries with the appropriate control switches. If the machine or any piece of equipment is found to be operational, locate the electrical circuit(s) supplying the power, and disconnect all such power sources. Then lock and tag these power sources.

i. Make sure the hydraulic system pressure is 0 PSI.

j. Before you begin any work on the machine or any auxiliary equipment, make sure that at least 60 seconds has elapsed since you turned off the main disconnect switch in step 'c'. (If a machine-mounted plate indicates that you should wait longer than 60 seconds, wait the recommended period of time before beginning any maintenance work.)

k. Verify that any equipment which may have been added, and which is not covered by steps 'a' - 'j' above, is considered for the lockout/tagout procedure.

l. After you have completed your work on the machine, make sure all guards, gates and other safety related devices are in place and functioning properly.

m. When the machine is completely ready to resume operation, remove your lock and tag from the main electrical disconnect switch. If someone else has placed a lock and/or tag on the main disconnect, do not remove the additional lock or tag. If there is no other lock or tag on the main disconnect, turn on the main disconnect switch and the electric motor, then perform the daily safety checks.

Power/Energy Isolation Devices

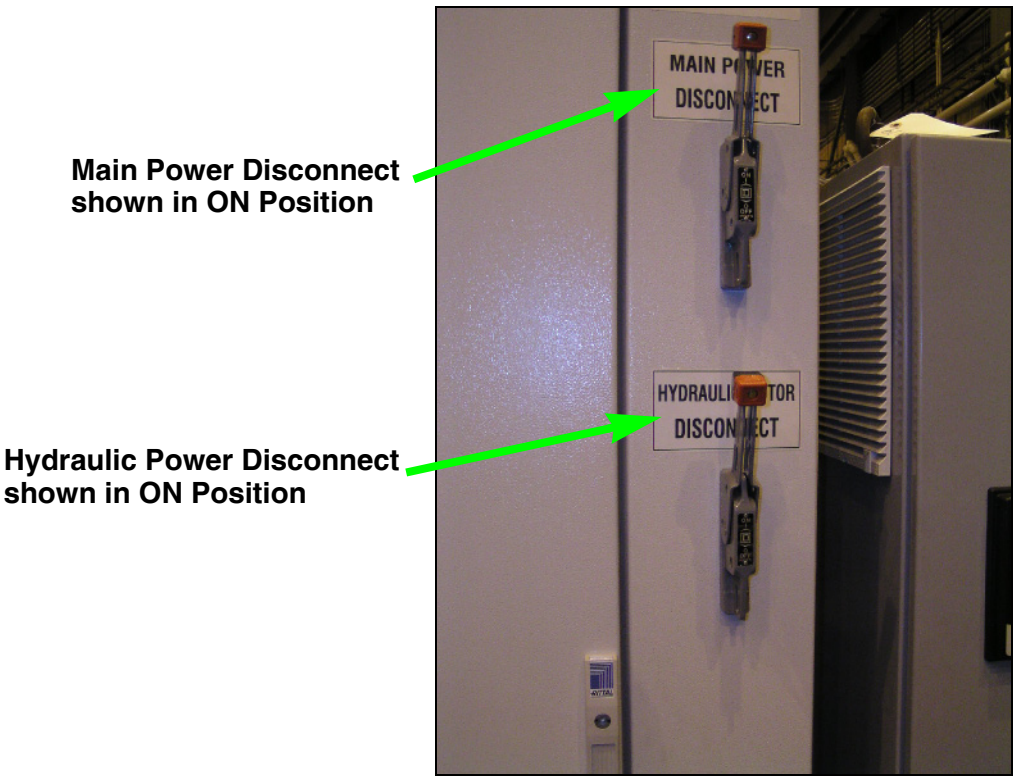
Electrical power to the machine can be locked/tagged out at the main disconnect located on the electrical cabinet. In addition, sectionalized disconnects for individual components of the machine (e.g. perform unscrambler, takeout devices) are located on their respective cabinets. When electrical power is turned off, residual energy remains for a period of time in the capacitors of the electrical system. Features built into the machine automatically deplete this residual energy. After turning off and locking out the disconnect, wait at least 60 seconds before beginning any maintenance or set up procedures.

When performing maintenance, service, set up, or troubleshooting, follow your company's written lock/tag out policies and procedures

Power/Energy Disconnect Locations

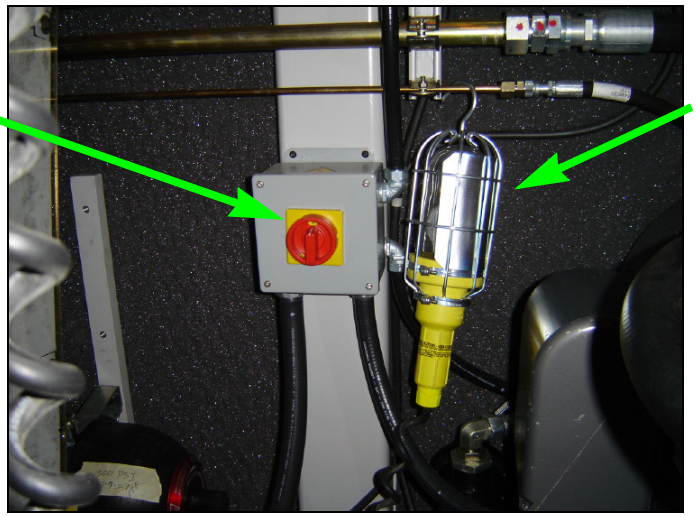
Lock/tag out capability is provided at three locations on the machine and downstream equipment:

- Main Power Disconnect shuts down power to the entire machine and downstream equipment.



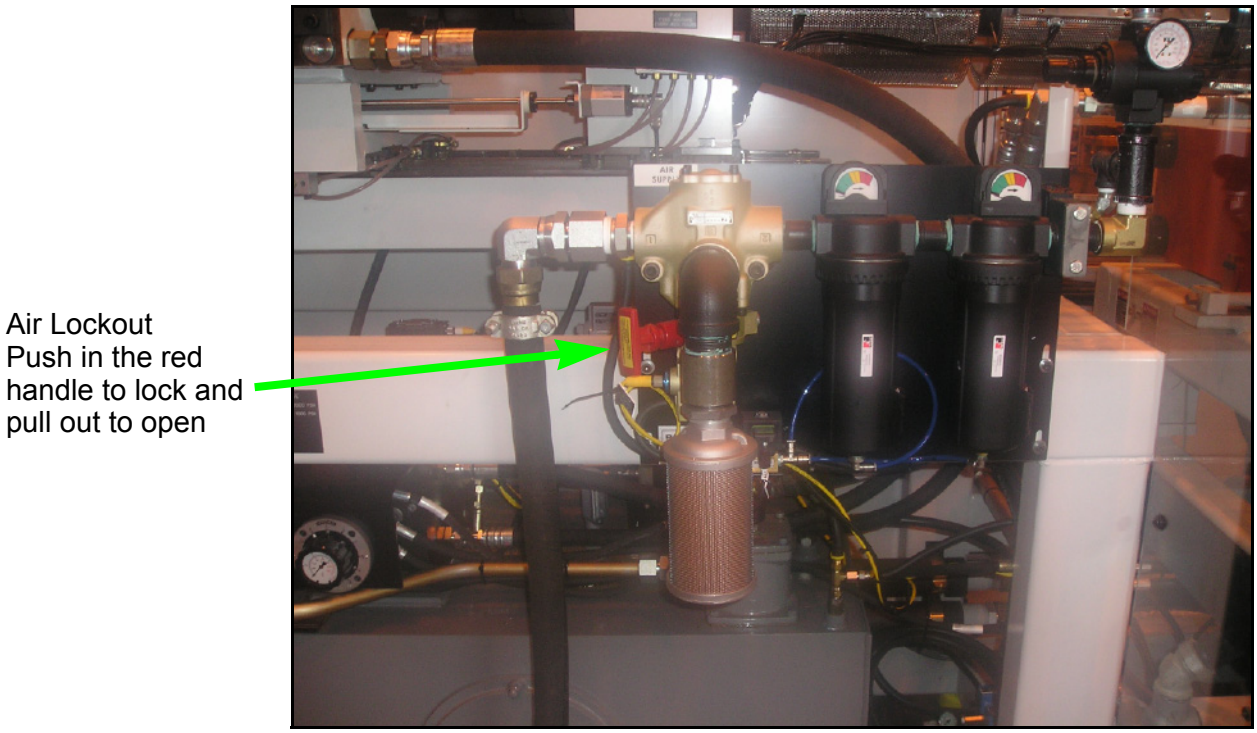
Power Disconnects

- Hydraulic Lockout - The hydraulic lockout is located below the Main Power Disconnect. As an option, it can be located inside the hydraulic area of the machine.



Internal Hydraulic Lockout

- Pneumatic (Air Lockout) - Push in the red handle to lock and pull out to open.



Air Lockout (UR 65 Shown)

Safety Systems and Devices

Following is a list of safety systems and devices on this blowmolding machine and auxiliary equipment (as applicable):

- **Emergency Stop Button**
  When depressed, this red mushroom-shape button shuts off all power to the machine except to the PLC.
- **Double-Ended Valving**
  When control circuits are energized, no sequence movement occurs.
- **Safety Signs**
  Safety signs used on this machine are approved ANSI or ISO safety signs for machinery.
- **Reliability Systems**
  This machine incorporates two independently controlled safety interlock systems. Prescribed safety relays monitor each system when opening and closing the safety gates. If a failure occurs in either system, the machine automatically shuts down and an alarm is activated. Any further cycling is prevented until the problem is corrected.
- **Energy-Isolating**
  Main (electrical and pneumatic) and sectionalized disconnects are provided at various locations on the machine. These disconnects require a user lock/tag out program for maintenance, service, troubleshooting, and set up. The main electrical disconnect shuts down all power except to the PLC. Sectionalized disconnects shut down power to specific component areas of the machine. The pneumatic disconnect dumps all pressure back to the tank.
- **Visual Warnings**
  Different alarms depict particular events. Visual alarms are various colors (blue, green, amber, and red) based on the severity or nature of the event.
- **Fixed Guards**
  Guards are held in place with a fastener. A tool is required to install or remove fixed guards.
- **Electrical Cycle Alarm Safety**
  Shuts down machine if automatic cycle time is too long.
- **Clamp Close Safety**
  Shuts down the machine if automatic cycle time is too long.
- **Low Pressure Mold Close Safety**
  Molds close under low pressure of approximately 24.5 bar (500 psi) and go to high pressure within 3.18 mm (.15 in.) of closing. If there is an obstruction after 1 to 1-1/2 seconds and the timer times out before the molds go into high pressure, the clamps open, the machine drops out of cycle, and an alarm is activated.
- **Mold Set Up Keyed Mode Selector**
  This key allows selection of dedicated manual mode. Travel is restricted to a maximum of 25 mm (1 in.) per second. This mode is controlled with a hold-to-run button (e.g. a job button). The key is only removable in set up mode.

Stack Lights

Blue: General Information - Light flashes on the Operator Console to alert the operator to a message or low-priority condition. The light flashes until the message or alarm is acknowledged and appropriate action is taken.

Green: Cycle Condition - Light flashes when the machine is in Auto or Manual Cycle mode to alert the operator that the machine is ready to cycle. When the machine is started in Auto mode, the light is steady. The light flashes when Cycle Stop is pressed during Auto mode. The light will not go on if the safety system has not been reset.

Amber: Warning Condition - Light flashes to alert the operator that the safety system circuit has been broken. The light will go off when appropriate corrective action has been taken and the safety circuit has been reset.

Red: Critical Condition - Light flashes when there is a high priority alarm condition (e.g. Emergency Stop). This condition shuts down the machine. The light continues to flash until appropriate corrective action is taken.

Audible Alarm

An audible alarm sounds when:

- Fault in the safety system is detected or a safety door is not closed.
- Auto Cycle is activated.
- Over temperature is too low or high.
- Cycle time is too long.
- Bottle Count is reached.

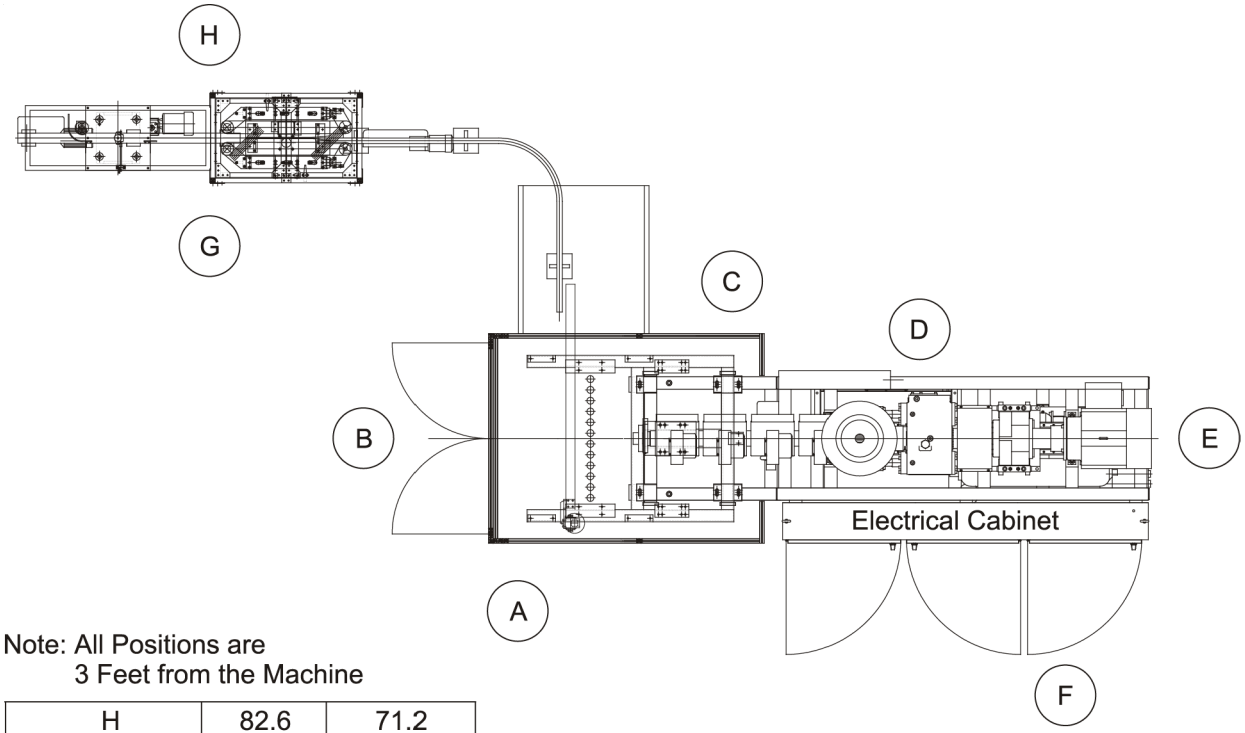
Safety Signs

For personal safety when operating the Uniloy Milacron machines, understand all warnings, mandatory prohibition, and lubrication signs.

Detailed information about the location of safety signs for your machine can be found in the machine layout drawings.

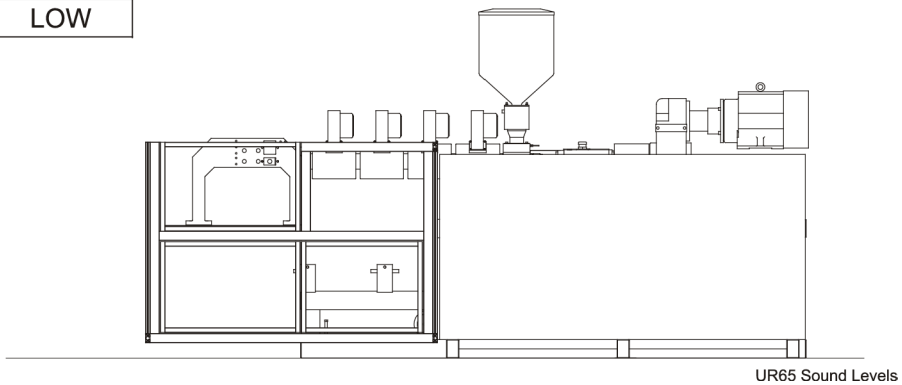
Sound Levels

The following diagram shows sound levels for the UR65 machine. For questions on the testing procedure or if more information is needed for your specific machine, please contact Uniloy. All operators and personnel working near machinery must wear the appropriate hearing protection per OSHA guidelines.



Note: All Positions are
3 Feet from the Machine

H	82.6	71.2
G	84.5	72.0
F	83.2	71.2
E	83.0	79.6
D	88.6	75.9
C	91.3	74.3
B	86.4	71.1
A	81.0	68.9
POSITION	PEAK	LOW



Sound Levels Near a UR65 Machine